Figure 3.7 shows an example data stream for ecommerce order logs generated and captured daily. It also shows that the order data stream consists of auto-generated hidden indexes by day. The data stream itself is nothing more than an alias for the time-series (rolling) hidden indexes behind the scenes. While search/read requests span all of the data stream's backing hidden indexes, indexing requests are only directed to the new (current) index.

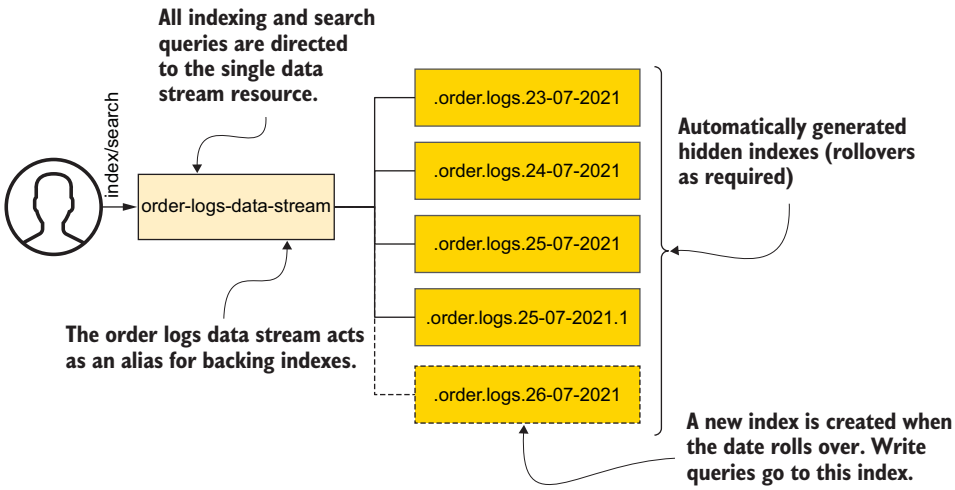


Figure 3.7 A data stream consists of automatically generated hidden indexes.

Data streams are created using a matching indexing template. *Templates* are blueprints consisting of settings and configuration values used for creating resources like indexes. Indexes created from a template inherit the settings defined in the template. We look at developing data streams with indexing templates in chapter 6.

In the last couple of sections, we briefly explained that indexes and data streams are distributed across shards and replicas. Let's dig deeper into these components.

3.2.4 Shards and replicas

Shards are software components that hold data, create support data structures (like inverted indexes), manage queries, and analyze data in Elasticsearch. They are instances of Apache Lucene allocated to an index during index creation. During the process of indexing, a document travels through to the shard. Shards create immutable file segments to hold the document in a durable file system.

Lucene is a high-performance engine for indexing documents efficiently. A lot goes on behind the scenes when indexing a document, and Lucene does this very efficiently. For example, documents are initially copied into an in-memory buffer on the shard and then written to writable segments before being merged and finalized to the